MDEIS

Simple Evaluation Methodology and Assessment Framework

1. Purpose

This document explains how MDEIS should evaluate educational information.

MDEIS uses a consistent process, but schools and other education providers can adjust the rules to suit their own needs. They may choose their own:

- Learning standards.
- Subjects and skills.
- Assessment methods.
- Scoring systems.
- Proficiency levels.
- Progress rules.
- Report formats.

MDEIS should provide clear, evidence-based findings. It should explain how it reached each conclusion and identify any limitations in the available information.

MDEIS must not make major decisions about a learner's future on its own. It should support teachers, schools, families and other authorised users.

2. Main principles

2.1 Use evidence

MDEIS must base its findings on available records, approved standards, assessment rules and clear calculations.

It must separate:

- Facts found in the records.
- Results calculated from those facts.
- Interpretations supported by the evidence.
- Possible explanations that need further investigation.
- Conclusions that cannot be made from the available information.

2.2 Allow local settings

Different schools may use different curricula, grading systems and learning standards.

MDEIS should allow each institution to set its own:

- Subjects and learning areas.
- Competencies and learning goals.
- Assessment categories.
- Assessment weights.
- Proficiency levels.
- Progress rules.
- Minimum evidence requirements.
- Report language and format.

Any changes to the settings should be recorded and applied consistently.

2.3 Consider the context

A score should always be considered in relation to:

- What the assessment measured.
- The learner's age or educational stage.
- The difficulty of the assessment.
- The assessment conditions.
- The standard used to interpret the result.

The same percentage from two different assessments may not represent the same level of learning in different circumstances.

2.4 Be clear and traceable

Users should be able to see where a finding came from.

Where possible, MDEIS should show:

- The assessment date.
- The source record.
- The calculation used.
- The standard or rule applied.
- The configuration version.
- Any important limitations.

2.5 Keep people involved

MDEIS should help people make better educational decisions. It may not replace professional judgment.

Teachers, school leaders, parents and other authorised users remain responsible for reviewing the findings and deciding what action is appropriate.

2.6 Support fairness

Assessment results can be affected by many factors, including:

- Access to learning resources.
- Language.
- Disability or accessibility needs.
- Previous curriculum exposure.
- Assessment conditions.
- Differences in teaching and learning opportunities.

A low score must not automatically be treated as proof of low ability, poor motivation or limited future potential.

3. Evaluation settings

Before analysing data, MDEIS must identify the settings that apply to the evaluation.

A school or authorised user may select an existing configuration or provide the information needed to create one.

3.1 Required settings

Setting	Purpose
Institution and educational context	Identifies the school, education system and learner stage.
Evaluation type	Shows whether the analysis is latitudinal or longitudinal.
Learner or group	Identifies the individual, class, cohort or group being assessed.
Subject or learning area	Identifies the subject, skill or competency being measured.

Learning standards	Defines the curriculum, learning goals or reference framework.
Assessment rules	Explains which assessments may be used and how they are scored.
Calculation rules	Explains how scores, averages, weights and changes are calculated.
Classification rules	Defines proficiency levels and their thresholds.
Comparison rules	Explains which results may be compared.
Reporting preferences	Defines the report style, language, charts and level of detail.
Evidence requirements	Defines the minimum information needed for a reliable conclusion.

3.2 Order of priority

When several standards or rules are available, MDEIS should use them in the following order:

1. Laws and official education regulations.
2. National or regional curriculum and assessment standards.
3. Approved school or institutional policies.
4. Relevant international frameworks.
5. User preferences that do not conflict with higher-priority rules.
6. General MDEIS defaults, used only when no other rule is available.

This is a proposed MDEIS rule. It is not a universal hierarchy set by the United Nations.

If two rules conflict, MDEIS should identify the conflict and ask for clarification or apply an approved solution. It must not combine conflicting rules without explanation.

3.3 Checking the settings

Before starting an evaluation, MDEIS should check that:

- The selected standards apply to the learners and educational stage.
- The scoring method is clearly defined.
- The proficiency levels are complete.
- Any assessment weights are valid.
- The reference documents are available.
- There is enough data for the requested analysis.
- The selected comparison method is suitable.

If important information is missing, MDEIS should ask for it, use a clearly stated safe default or provide a limited analysis explaining what cannot be confirmed.

4. Standard evaluation process

MDEIS should follow the steps below for every evaluation.

Step 1: Understand the request

The system should identify:

- What the user wants to know.
- Which learner or group is involved.
- Which subjects or skills are relevant.
- The period being examined.
- The type of analysis, longitudinal or cross-sectional, requested.
- Who will receive the report

The system should distinguish between requests to:

- Measure proficiency.
- Compare results.
- Track progress.
- Identify learning gaps.
- Examine group patterns.
- Suggest areas for human review.

Step 2: Find the relevant information

MDEIS should collect the applicable:

- Assessment records.
- Learning standards.
- Calculation rules.

- Classification rules.
- Reporting settings.

It should record which sources were actually used.

MDEIS must not claim to follow a framework simply because that framework is stored in its reference library.

Step 3: Check the data

The system should look for:

- Missing test results.
- Duplicate records.
- Conflicting records.
- Invalid scores.
- Incorrect denominators.
- Inconsistent learner names or identification numbers.
- Missing dates.
- Missing educational-stage information.
- Changes in grading systems.
- Differences in assessment difficulty or content.
- Too little evidence for the requested conclusion.

A missing result must not automatically be treated as zero.

If data problems affect the result, MDEIS should explain the problem and limit its conclusions.

Step 4: Link results to learning goals

Each assessment result should be linked to the relevant:

- Subject.
- Learning area.
- Competency.
- Learning objective.

If an assessment covers several skills, MDEIS should use item-level results, an approved marking guide or an authorised mapping.

If only an overall examination mark is available, MDEIS must not invent results for individual topics.

Step 5: Calculate the results

MDEIS should apply the approved mathematical and statistical rules.

Where possible, calculations should be completed by reliable software tools rather than by the language model alone.

The system should keep enough information for the calculations to be checked or repeated.

Step 6: Apply proficiency levels

The results should be interpreted using the approved thresholds and descriptions.

If no suitable classification system is available, MDEIS may report the numerical results and use careful descriptive language.

It must not present invented thresholds as official standards.

Step 7: Explain the findings

MDEIS should explain:

- What the results show.
- Which evidence supports the findings.
- What the evidence does not show.
- Which explanations are possible but unconfirmed.

The system should not claim to know the cause of a result unless there is evidence to support that claim.

Step 8: Prepare the report

The report should include the requested findings, calculations, classifications, graphs, limitations and sources where relevant.

It may also include questions for teacher review or suggestions for further assessment.

Step 9: Human review

Before the findings are used for an important educational decision, an authorised person (admin) should review the evidence and limitations.

The system should record, where appropriate:

- The evaluation date.
- The type of request.
- The configuration version.
- The data sources.

- The type of analysis
- The calculation method.
- The findings.
- The review status.

This information must be handled according to privacy and data-retention requirements.

5. Evaluation type A: Cross-sectional analysis

5.1 Purpose

Cross-sectional analysis looks at the learning performance of a group at one point in time, over a number of years or during one assessment period

It can help a school understand:

- How many learners meet a learning standard.
- Which learning areas are strong.
- Which areas may need support.
- Whether further assessment is needed.

This type of analysis focuses on group patterns. It should not rank individual learners unless a separate individual analysis has been requested and approved.

5.2 Possible evidence (Data sets to reference from)

Depending on the school's settings, evidence may include in-built diagnostic assessments found in MDEIS which could test reading, writing, numeracy, spatial reasoning and a host of other relevant fields. Evidence may also come from classroom tests from a particular institution.

Each assessment should have clear information about:

- Its source.
- Its purpose.
- The age or grade it is designed for.
- The language used.
- The scoring method.

A short diagnostic quiz should not automatically be treated as equal to a validated standardised assessment or a high-stakes examination.

5.3 Main measures

MDEIS may calculate the following.

A. Proficiency distribution

This shows the percentage of learners in each approved proficiency level.

$P_k = n_k / N \times 100\%$

Where:

- (P_k) is the percentage of learners in level (k).
- (n_k) is the number of learners in level (k).
- (N) is the number of learners with valid results.

For example, suppose a class of 40 learners has 12 learners classified as proficient in a particular learning area, then $P_k = 12/40 \times 100 = 30\%$

B. Performance by learning area

This shows results for areas such as:

- Reading comprehension.
- Written expression.
- Numeracy.
- Mathematical reasoning.

C. Meeting a threshold

This shows the percentage of learners who meet a defined learning expectation.

The threshold must be suitable for the assessment and the learning standard.

D. Evidence coverage

This shows how many learners have usable results for each learning area.

It should also show where missing results limit the findings.

E. Differences between learning areas

This shows how results differ across subjects or skills.

These differences should be interpreted carefully if the assessments use different scales, levels of difficulty or groups of learners.

5.4 Interpretation rules

MDEIS should:

- Report group patterns rather than make unsupported claims about individual students.
- Show the number of valid results behind each percentage.
- Separate the average score from the percentage meeting a standard.

- Identify areas where many learners may need support.
- Identify areas where learners show strong performance.
- Avoid treating one assessment as a complete picture of learning.
- Consider access, language and assessment conditions.
- Avoid blaming teachers, schools or curricula without additional evidence.

5.5 Example

A group assessment may show that most learners meet the numeracy threshold, while many learners fall below the reading-comprehension threshold.

MDEIS should report:

- The percentages.
- The number of learners assessed.
- The thresholds used.
- The assessment sources.
- Any important limitations.

It may suggest further reading assessment or instructional review.

It should not automatically conclude that the school's teaching methods caused the result.

6. Evaluation type B: Longitudinal analysis

6.1 Purpose

Longitudinal analysis looks at how an individual learner's results change over time.

The period may include:

- Several assessments.
- Different terms.
- Multiple school years.
- Different educational stages.

The purpose is to describe observed progress, identify patterns and help teachers investigate possible learning needs.

6.2 Possible evidence

Evidence (data) may include:

- Examination results.
- Termly assessments.
- Student report cards(annual results).
- Sequence test scores.
- Assignments andProjects where present and applicable
- Attendance records, where relevant and authorised.
- Teacher observations.

- Learning-support records.

These types of evidence should not be treated as equal.

For example, attendance may provide useful context, but it does not directly prove subject proficiency.

6.3 Main measures

A. Current performance

This is the learner's most recent valid result, together with:

- The date.
- The subject or skill.
- The assessment type.
- The assessment conditions.

B. Absolute score change

$\Delta S = Sr - Sp$

Where ΔS is the absolute score change, Sp is the earliest available score and Sr is the most recent score

This shows the difference between two results on a comparable scale. A relative percentage change may also be calculated where necessary.

C. Change in proficiency level

This shows whether the learner moved between approved proficiency levels.

The same or a clearly comparable classification system should be used.

D. Consistency

This shows how much the learner's results vary over time.

Assessment differences should be considered before interpreting this variation.

E. Topic-level progress

This shows changes in specific topics or competencies when enough detailed evidence is available.

F. Evidence strength

This considers:

- The number of assessments.
- The dates of the assessments.
- The time period.
- Whether the assessments can be compared.

All of these measures may be plotted in a graph of score against time to ease visualization of results.

6.4 Trend labels

When there is enough evidence, MDEIS may use the following labels:

- **Improving:** Results show an upward pattern under the approved trend rule.
- **Relatively stable:** Results remain within the approved stability range.
- **Declining:** Results show a downward pattern under the approved trend rule.
- **Mixed:** Different subjects or skills show different patterns.
- **Insufficient evidence:** There is not enough reliable information to identify a trend.

The configuration must define:

- The number of results needed.
- The time period.
- The size of change required.
- The stability range.

MDEIS must not label a learner as improving or declining based only on two results from assessments that cannot reasonably be compared.

6.5 Interpretation rules

MDEIS should:

- Compare results only when the assessments are reasonably comparable.
- Note changes in curriculum, grading or assessment methods.
- Separate a change in marks from a confirmed change in learning.
- Report topic-level strengths only when detailed evidence is available.
- Treat possible explanations as unconfirmed unless supported by evidence.
- Avoid using short-term results to predict a learner's entire future.

6.6 Example

A learner's Mathematics results may rise across three comparable assessments, while English results fall during the same period.

MDEIS should report:

- The scores.
- The dates.
- The assessment conditions.
- The calculations used.

It may suggest further review of the English results.

It should not automatically conclude that the learner has lost interest, lacks ability or has received poor teaching.

7. Proficiency levels

MDEIS should allow each institution to use its own proficiency levels.

The school must define what each level means and how it is assigned.

A simple example is shown below.

Level	Description
Beginning	The learner has not yet shown the key learning expectations consistently.
Developing	The learner shows some of the expected skills but still needs support in certain areas.
Proficient	The learner meets the expected standard for the relevant educational stage.
Advanced	The learner performs above the expected standard in the assessed area.
Insufficient evidence	There is not enough reliable information to assign a level.

These are suggested descriptions. They are not official UNESCO categories.

A school should define the learning outcomes and score ranges linked to each level.

A percentage score should not be used as the only basis for proficiency unless the assessment and classification rules support that use.

Where several skills are assessed, MDEIS should show the learner's full profile rather than reduce everything to one label.

8. Assessment weighting

Schools may use different weights for different assessments.

For example, a school may give different importance to:

- Examinations.
- Practical work.
- Assignments.
- Projects.
- Quizzes.

MDEIS must check that:

- The weights are clearly defined.
- The scores can reasonably be combined.
- Missing results are handled according to an approved rule.

Missing assessments must not automatically be treated as zero.

The report should show how the final score was calculated and, where possible, retain the individual assessment results.

A weighting system does not automatically prove that the final score measures every important skill accurately.

9. Changing institutional settings

The general MDEIS method should be kept separate from each institution's own rules.

This allows the process to remain consistent while schools change their:

- Standards.
- Subjects.
- Assessment rules.
- Proficiency levels.
- Reporting preferences.

Each configuration should include:

- A unique identification number (matricule).
- The institution or context where it applies.
- The subjects and educational stages covered.
- The source and version of each standard.
- The scoring and classification rules.
- The start date.
- The review or expiry date, where relevant.

- The person or body that approved it.
- A record of changes.

When a user submits a request, MDEIS should identify the correct configuration and check that it is authorised for the task.

If a user asks for different thresholds or criteria, MDEIS should determine whether this is:

- A temporary analysis request.
- A formal change to the institution's approved rules.

A user request must not silently change permanent institutional settings.

If a change affects comparisons over time, MDEIS should explain the change and, where possible, show results under both the old and new settings.

10. Evidence and uncertainty

Each important report should include, where relevant:

1. **Question:** What the user asked MDEIS to examine.
2. **Scope:** The learner or group, subjects and period covered.
3. **Evidence:** The records and standards used.
4. **Method:** The calculations and rules applied.
5. **Findings:** What the evidence directly shows.
6. **Interpretation:** What the findings may mean.
7. **Limitations:** Missing data, assessment differences and other concerns.
8. **Follow-up:** Questions or further assessments for human review.

MDEIS should clearly separate:

- A calculated result.
- An interpretation produced by the AI.
- A decision made by a human.

If the evidence is incomplete, conflicting or unclear, MDEIS should say so instead of giving a false sense of certainty.

11. Safeguards

MDEIS must not:

- Invent results, standards, thresholds or sources.
- Present guesses about causes as facts.
- Infer intelligence, motivation, character or future potential from scores alone.
- Make independent decisions about admissions, promotion, exclusion, discipline or careers.
- Treat missing records as poor performance.

- Compare results from incompatible assessment systems without a suitable method.
- Treat group results as proof of an individual's performance.
- Share confidential learner information with unauthorised users.
- Change approved institutional settings without permission.

If the results suggest a serious learning difficulty, MDEIS should present the evidence and recommend human review. It should not make a diagnosis.

12. Human review and responsibility

An authorised person should be able to review, question and correct any MDEIS evaluation.

The reviewer should be able to see:

- The records used.
- The rules applied.
- The calculations made.
- The reasons for the findings.

Where necessary, the reviewer should be able to:

- Request a recalculation.
- Add new evidence.
- Correct an error.
- Record a different interpretation.

The system should distinguish between:

- A draft produced by MDEIS.
- A report reviewed by an authorised person.
- A corrected or replaced report.
- A decision made by a responsible human.

Where appropriate and lawful, the system should record:

- The reviewer's identity.
- The review date.
- Any changes made.

A report must not suggest that a human approved it if no human review took place.

13. Reference framework

MDEIS should maintain a library of:

- Official education standards.
- Assessment guidance.
- Institutional policies.

- Relevant international frameworks.
- Technical documents on responsible AI.

The following sources may support the development of MDEIS.

13.1 United Nations Sustainable Development Goal 4

Source: United Nations Department of Economic and Social Affairs. *Goal 4: Quality Education.*

Relevance: SDG 4 supports inclusive and equitable quality education. Target 4.1 and indicator 4.1.1 refer to minimum proficiency in reading and mathematics at certain educational stages.

Use in MDEIS: It may support broad policy alignment and monitoring of foundational learning.

It should not be treated as a universal scoring system for every subject or learner.

13.2 UNESCO Institute for Statistics: Global Proficiency Framework

Source: UNESCO Institute for Statistics. *Global Proficiency Frameworks and related learning-assessment resources*.

Links:

- https://www.uis.unesco.org/en/gaml/monitoring-framework
- https://www.uis.unesco.org/en/publication/global-proficiency-framework-gpf-mathematics
- https://learningdata.uis.unesco.org/index.php/resources/methods/

Relevance: The framework describes knowledge and skills expected at certain grade levels in reading and mathematics.

Use in MDEIS: Relevant parts may help with:

- Foundational literacy and numeracy.
- Learning-domain mapping.
- Proficiency descriptions.
- Interpretation of assessment results.

MDEIS must check that the framework applies to the learner's stage, subject and intended use.

13.3 UNESCO Institute for Statistics: Principles of Good Practice in Learning Assessments

Source: UNESCO Institute for Statistics. *Principles of Good Practice in Learning Assessments.*

Relevance: This resource provides guidance on the quality and use of learning assessments.

Use in MDEIS: It may help guide assessment quality, data interpretation, reporting limitations and assessment design.

13.4 UNESCO: AI Competency Framework for Teachers

Source: Miao, F., & Cukurova, M. (2024). *AI competency framework for teachers.* UNESCO.

Relevance: The framework focuses on human-centred AI, ethics, teacher responsibility and responsible use of AI in education.

Use in MDEIS: It may support the design of:

- Human review.
- Teacher responsibility.
- Transparent workflows.
- Ethical use of AI.

It is not a student-performance scoring standard.

13.5 UNESCO: AI Competency Framework for Students

Source: Miao, F., Shiohira, K., & Lao, N. (2024). *AI competency framework for students.* UNESCO.

Relevance: The framework focuses on human agency, ethical AI and responsible participation in AI-supported environments.

Use in MDEIS: It may support the system's approach to learner agency, transparency and responsible technology use.

It is not a general framework for measuring reading, mathematics or other school subjects.

13.6 UNESCO: Guidance for Generative AI in Education and Research

Source: UNESCO (2023). *Guidance for generative AI in education and research*.

Relevance: This guidance addresses human agency, privacy, age-appropriate use and responsible AI in education.

Use in MDEIS: It may help shape:

- Privacy safeguards.
- Responsible-use policies.
- Human review.
- The handling of sensitive learner information.

14. Quality checks

Before MDEIS is used in practice, it should be tested with realistic examples and known expected results.

Testing should cover:

- Correct score calculations.
- Correct progress calculations.
- Correct use of proficiency thresholds.
- Missing and inconsistent data.
- Assessments that cannot be compared.
- Cross-sectional and longitudinal analysis.
- Insufficient evidence.
- Correct use of reference standards.
- Unsupported claims about causes.
- Unauthorised configuration changes.
- Privacy-sensitive information.
- Repeated use of the same data and settings.
- Conflicting instructions and reference documents.

Tests should include:

- Normal cases.
- Difficult cases.
- Ambiguous requests.
- Incomplete records.
- Incorrect data.

The results of testing should be recorded, including errors and unresolved problems.

A report should not be considered reliable simply because it sounds convincing.

15. Expected report content

Every MDEIS evaluation should include, where relevant:

- The evaluation type.
- The learner or group covered.
- The period examined.
- The standards and configuration used.
- The calculations completed.
- The findings.
- The proficiency levels and their meanings.
- The evidence supporting the findings.
- The limitations and uncertainty.
- Suggested follow-up for human review.

Reports may include:

- Tables.
- Graphs.
- Percentages.
- Proficiency distributions.
- Progress timelines.
- Learning profiles.

The format may differ between institutions, but the basic requirements for accuracy, evidence, transparency and responsible interpretation should remain the same.

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16. Conclusion

MDEIS is designed to turn educational records into clear and useful information about learning.

It combines a consistent evaluation process with settings that can be adapted to different schools and education systems.

It can support:

- Group-level reviews of foundational learning.
- Individual progress tracking.
- Identification of possible learning gaps.
- Better questions for teachers and schools.
- More informed educational planning.

MDEIS should not place permanent labels on learners or make important decisions without human involvement.

The framework should be reviewed by educators, assessment specialists, schools and policy experts before it is used as a formal operational standard.

Core commitment: MDEIS will measure only what the available evidence and approved criteria support. It will explain how it reaches its findings, state what it cannot confirm and leave important educational decisions to responsible human beings.